Dataset and benchmark limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which source and setting each denominator represents.

Teach benchmark limitations through explicit grouping and missing coverage.

Keep one habit: Model eight-source count → distinguish identity from independence → trace indoor/outdoor slices → preserve unassessed fields → assess independent source arrangement.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of dataset and **benchmark** limitations

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of sensitivity analysis

Reference example, not the only acceptable wording: Teacher explanation: Sensitivity analysis checks whether a conclusion changes under plausible settings or analysis choices.

It should not secretly select the most favourable result. Prespecify a range and preserve all results. Distinguish robustness across settings from proof beyond the tested range.

Settings affecting data roles require special caution.

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of multiple comparisons

Reference example, not the only acceptable wording: Teacher explanation: Testing many hypotheses creates more chances for false discoveries. A family rule must be specified before selecting attractive p-values.

For this exercise use Bonferroni:family alpha .05 divided by number of tests m; reject each test when its valid p-value is at most .05/m. This controls the family error by a bound without requiring independent tests.

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of uncertainty intervals

Reference example, not the only acceptable wording: Teacher explanation: An uncertainty interval needs a sampling model and assumptions.

A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96:mean plus/minus1.96*sigma/sqrt(n).

Unknown sigma requires a different justified method.

Pictured end goal

Explain dataset and **benchmark** limitations using the records and boundaries of this fictional investigation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Sensitivity analysis

Multiple comparisons

Uncertainty intervals

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. A **benchmark** measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of dataset and **benchmark** limitations

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach benchmark identity and coverage with honest denominators

> Distinguish unique IDs, independent sampling and representative evaluation.

Definitions and full contract

A **benchmark** measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

Audit item	Complete supplied contract
Main source arrangement	Ten rows: three copies of S1 plus seven other distinct source IDs S2–S8
Distinct source count	One repeated source + seven others = eight
Duplicate weighting	S1 contributes three rows; each other source contributes one
Extra rows beyond one per source	Two; do not confuse three copies with three extra copies
Independence	Different source IDs do not prove independent sampling
Separate picture	Indoor 9/10; outdoor initially unmeasured

Audit item	Complete supplied contract
Changed picture	Four reserved outdoor cases supplied, two correct
Unassessed properties	Source independence, label validity, eligibility, future representativeness and raw per-source outcomes

Main source ledger and checked teacher reasoning

Declared source ID	Row IDs in illustrative arrangement	Row weight
S1	R1, R2, R3	3 of 10
S2	R4	1 of 10
S3	R5	1 of 10
S4	R6	1 of 10
S5	R7	1 of 10
S6	R8	1 of 10
S7	R9	1 of 10
S8	R10	1 of 10
Total	Ten rows; eight distinct IDs	S1 receives triple row weight
Evidence limit	IDs illustrate supplied arrangement	No labels or independent sampling inferred

Main: ten rows consist of three copies of one source plus seven other distinct sources, giving eight source IDs and two extra rows beyond one per source. Do not call those eight statistically independent merely because IDs differ. No per-source labels are supplied, so no deduplicated accuracy can be calculated. Picture: indoor 9/10 = 90%; original outdoor performance is unknown.

Added outdoor 2/4 = 50%; pooled 11/14 ≈ 78.6% is a different quantity. Independent twelve-row case: four copies of one source plus eight others give nine distinct source units and three extra rows. A valid coverage concern names an unassessed setting, source population or

label property without pretending it has been measured. No actual photo study or population certainty follows.

Dataset and benchmark limitations — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Ten benchmark rows include three copies of one source and seven other unique sources.	Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.	Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Ten **benchmark** rows include three copies of one source and seven other unique sources.

Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Identity and coverage probes

Benchmark audit	Checked interpretation
Three copies of one source plus seven others	Eight source IDs, not ten independent sources
Three copies	Two extra rows beyond one copy, not three extra rows
Source IDs differ	Identity distinction does not prove sampling independence
No outcome labels supplied per source	Do not calculate an invented deduplicated accuracy
Exclude duplicates after inspecting outcomes	Disclose analysis change; preserve original results and declared unit rule
No outdoor cases	Unknown performance, not zero accuracy
Outdoor 2/4 and pooled 11/14	Different denominators and target descriptions
High indoor result	No automatic outdoor-fitness conclusion
Observed slice difference	Does not by itself establish cause or significance

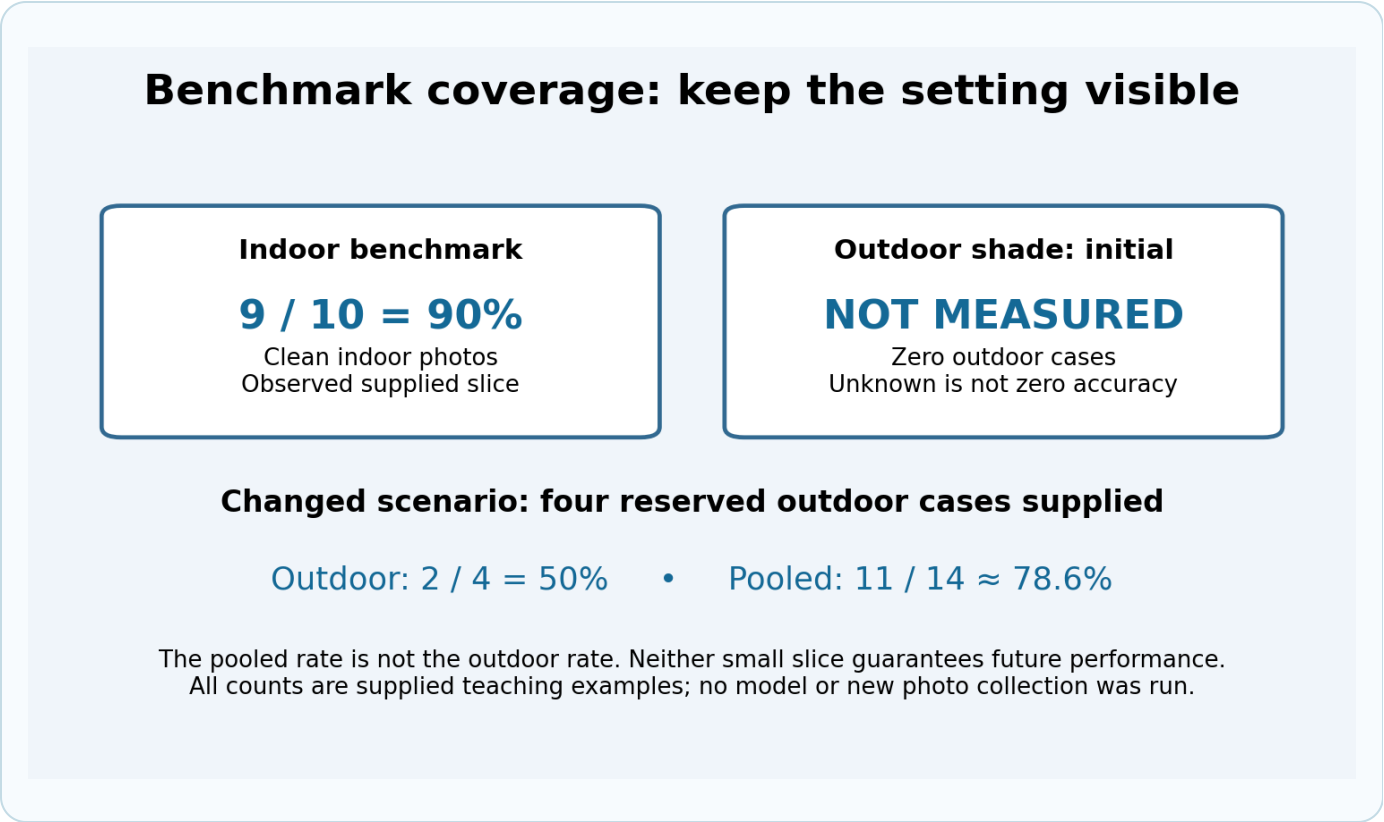
Separate photo settings and native coverage chart

A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos.

Indoor **benchmark**: 9/10 correct. No outdoor cases are present. The same model is intended for both.

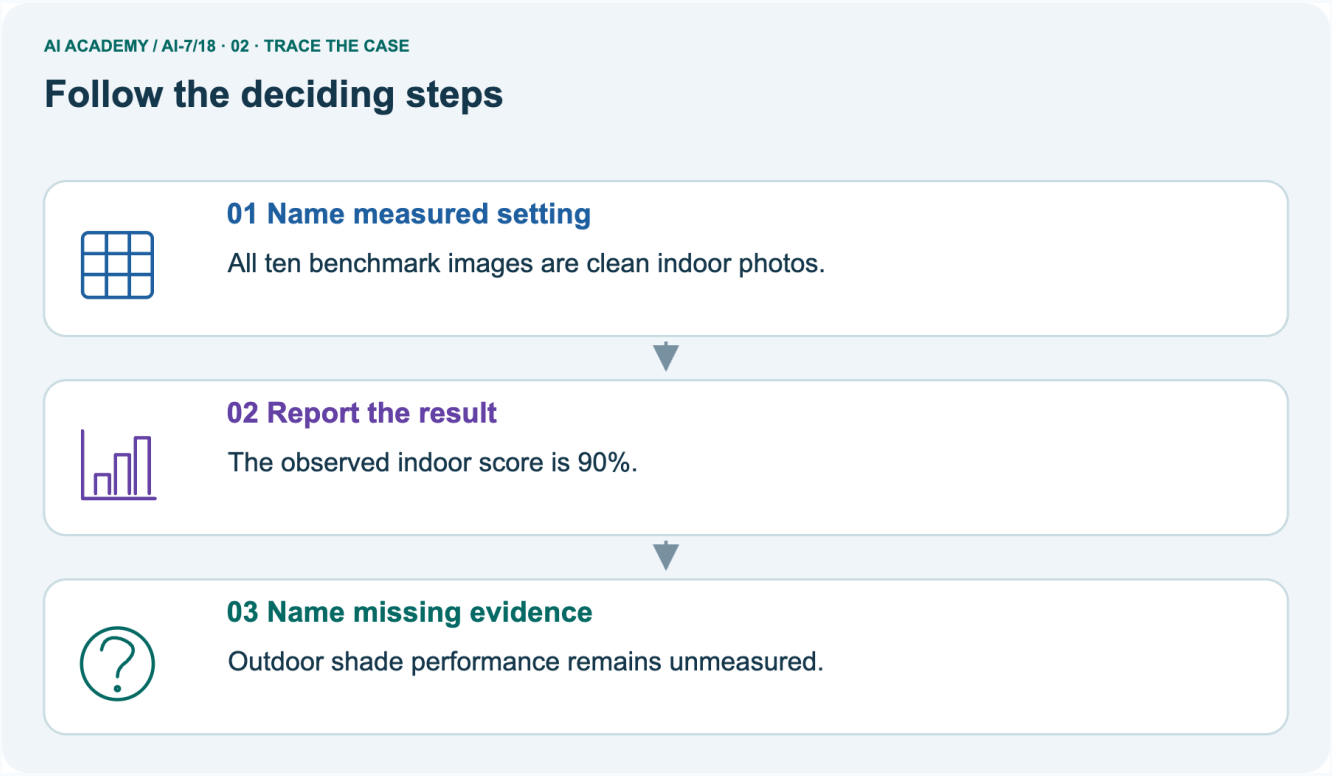
The picture's original ten photos are all clean indoor images: nine correct gives 90% on that supplied slice. There are no outdoor cases, so outdoor accuracy is unmeasured. In the changed teaching scenario, four reserved outdoor cases are supplied and two are correct, giving 50% for that slice. Pooling both settings gives 11/14, approximately 78.6%, but that pooled rate is not outdoor performance.

The lower observed outdoor slice does not identify the cause, establish significance or prove all future outdoor images will behave similarly. No new photos or model outputs were actually collected for this lesson.



Setting	Original supplied evidence	Changed supplied evidence
Indoor	9/10 = 90%	Unchanged 9/10 = 90%
Outdoor shade	No cases; rate unknown	2/4 = 50%
Pooled descriptive rate	9/10, entirely indoor	11/14 ≈ 78.6%
Appropriate outdoor report	Unmeasured	2/4 in the supplied outdoor slice
Unsupported extension	Assume outdoor also 90%	Use pooled 78.6% as outdoor accuracy
Remaining limits	Coverage absent	Small slice, sampling and labels still need inspection

Picture trace and changed outdoor key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name measured setting	All ten benchmark images are clean indoor photos.
2. Report the result	The observed indoor score is 90%.
3. Name missing evidence	Outdoor shade performance remains unmeasured.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

Neither slice estimates all future photos; inspect sampling, labels and grouping.

Change: Add four reserved outdoor cases, of which two are correct.

Prediction: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Benchmark	A declared task dataset and scoring procedure
Duplicate unit	Repeated representation of the same underlying entity
Coverage gap	A relevant condition absent from evaluated data

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Dataset and benchmark limitations

The facts we will use

AI ACADEMY / AI-7/18

Benchmark conditions

	Correct	Total
Indoor benchmark	9	10
Added outdoor slice	2	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/18

Dataset and benchmark limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name measured setting

All ten benchmark images are clean indoor photos.



2. Report the result

The observed indoor score is 90%.



3. Name missing evidence

Outdoor shade performance remains unmeasured.

CHANGE ONE THING

Add four reserved outdoor cases, of which two are correct.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Dataset and **benchmark** limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Benchmark** success does not establish performance on a deployment population missing from its design.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Report the indoor **benchmark** with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

All ten **benchmark** images are clean indoor photos. The observed indoor score is 90%. Outdoor shade performance remains unmeasured. Changed case: Report the outdoor slice as 2/4=50%, separately from indoor results.

The new records finally measure the missing condition, with a small denominator.

Changed-case reasoning: Report the outdoor slice as 2/4=50%, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main ten rows have eight source IDs, with a repeated-source weight of three and two extra rows. Picture indoor 9/10 and added outdoor 2/4 use different slice denominators; pooled 11/14 is not outdoor accuracy. Independent twelve rows with four copies of one source plus eight others give nine source IDs and three extra rows.

Require a concrete coverage concern, without asserting source independence or inventing missing labels/outcomes.

Diagnose the current misconceptions

The worked result establishes every future case: **Benchmark** success does not establish performance on a deployment population missing from its design.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

Teacher explanation: Nine unique units. Check whether cases represent requested time, groups and tasks; row count alone does not answer coverage.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: A **benchmark** measures performance on its particular data and labels. Duplicates, obsolete cases or poor coverage can weaken claims.

Inspect eligibility, source dates, related units and label definitions before scoring. Deduplicate by declared unit identity, not merely file name; report excluded and unassessed cases.

W2 Trace the worked case

Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

W3 Audit the limit

A report omits this qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: **Benchmark** success does not establish performance on a deployment population missing from its design.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Benchmark** success does not establish performance on a deployment population missing from its design.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Qualification: **Benchmark** success does not establish performance on a deployment population missing from its design.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Twelve rows include four copies of one source and eight others. Count unique units and name one coverage concern.

Teacher explanation: Nine unique units. Check whether cases represent requested time, groups and tasks; row count alone does not answer coverage.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Nine unique units. Check whether cases represent requested time, groups and tasks; row count alone does not answer coverage. Require intermediate reasoning and the stated scope.

Fresh retry: Eight rows include two copies of a single source and six others. Expected reasoning: Seven unique units. Report duplicate policy and original row inventory.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled dataset and **benchmark** limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of dataset and **benchmark** limitations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Research ethics and consent. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/19, ai-7/21, ai-7/24. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of dataset and **benchmark** limitations

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Ten **benchmark** rows include three copies of one source and seven other unique sources. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Eight unique source units, not ten independent sources. Identify duplicate weighting before analysis; do not silently discard results.

Check every learner before discussion: Use the worked case for Dataset and **benchmark** limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Benchmark** success does not establish performance on a deployment population missing from its design.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add four reserved outdoor cases, of which two are correct. Require a **prediction** and a reason before comparing. Expected result: Report the outdoor slice as 2/4=50%, separately from indoor results.

Reason: The new records finally measure the missing condition, with a small denominator.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

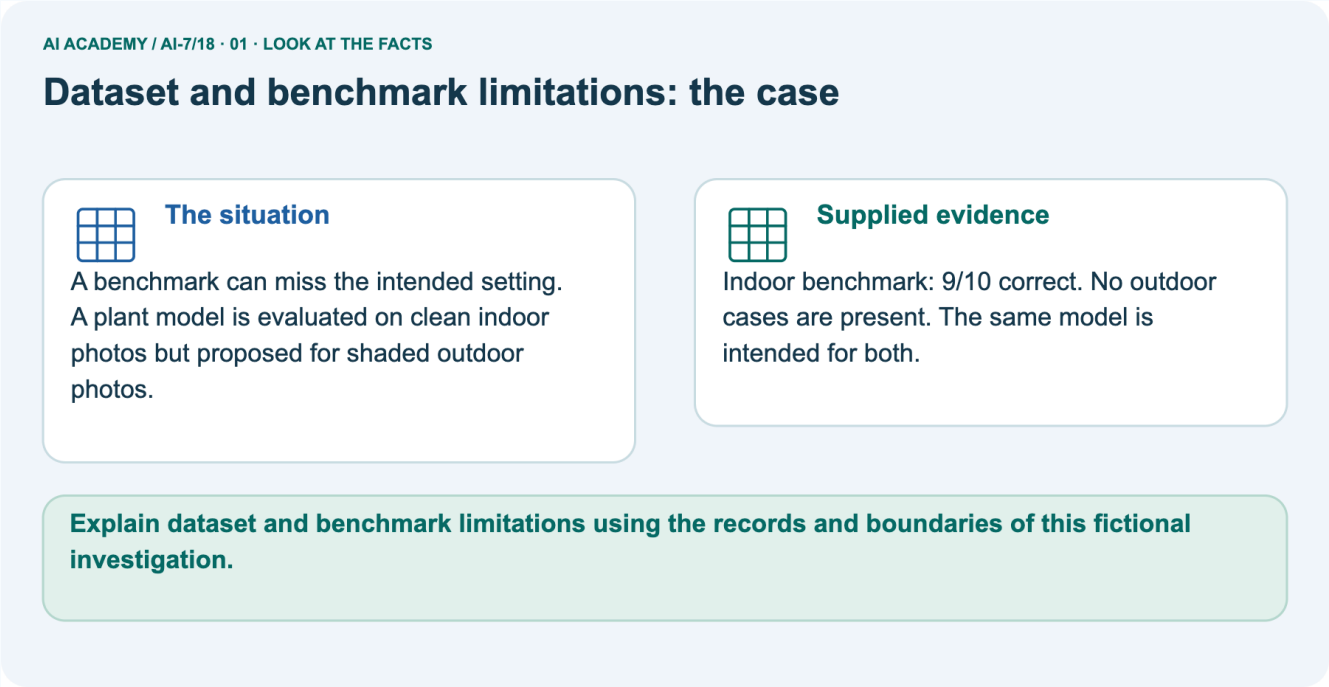
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Dataset and benchmark limitations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

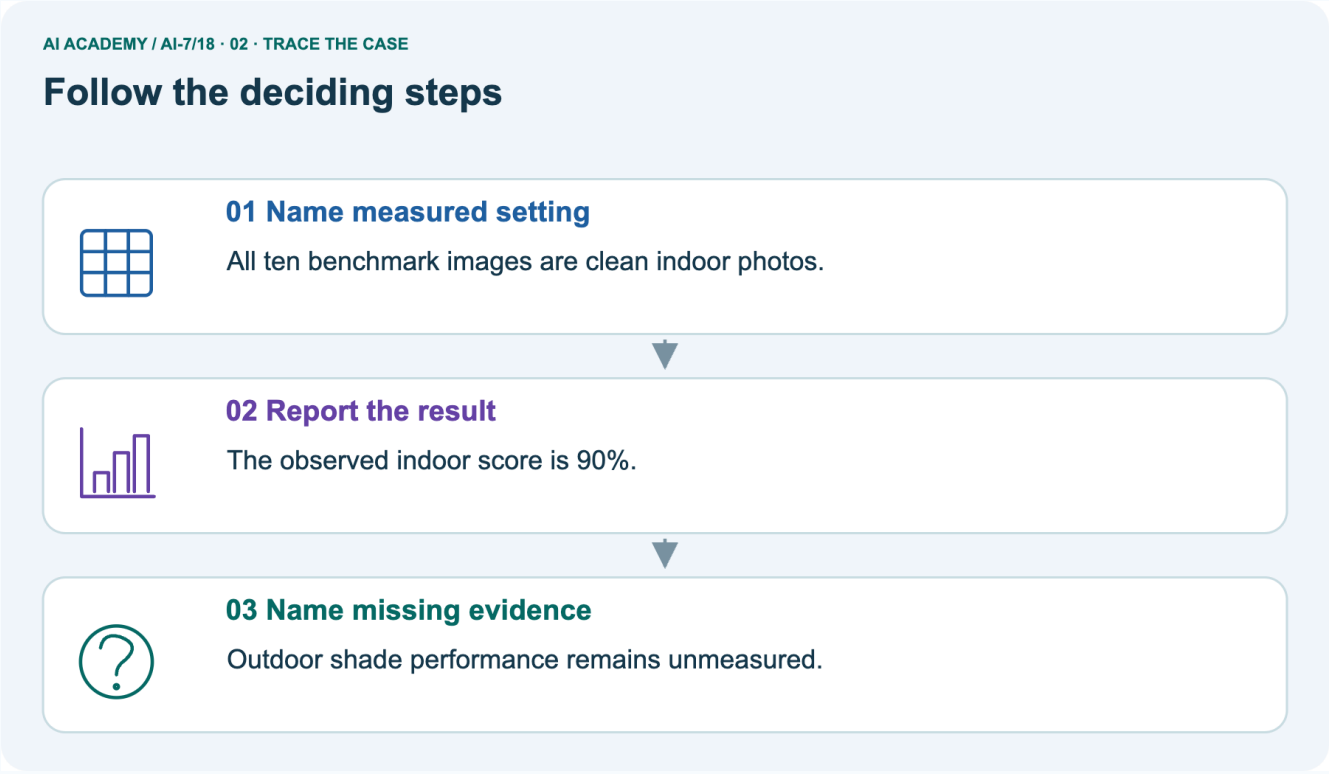
A **benchmark** can miss the intended setting. A plant model is evaluated on clean indoor photos but proposed for shaded outdoor photos. Supplied facts: Indoor **benchmark**: 9/10 correct. No outdoor cases are present.

The same model is intended for both.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



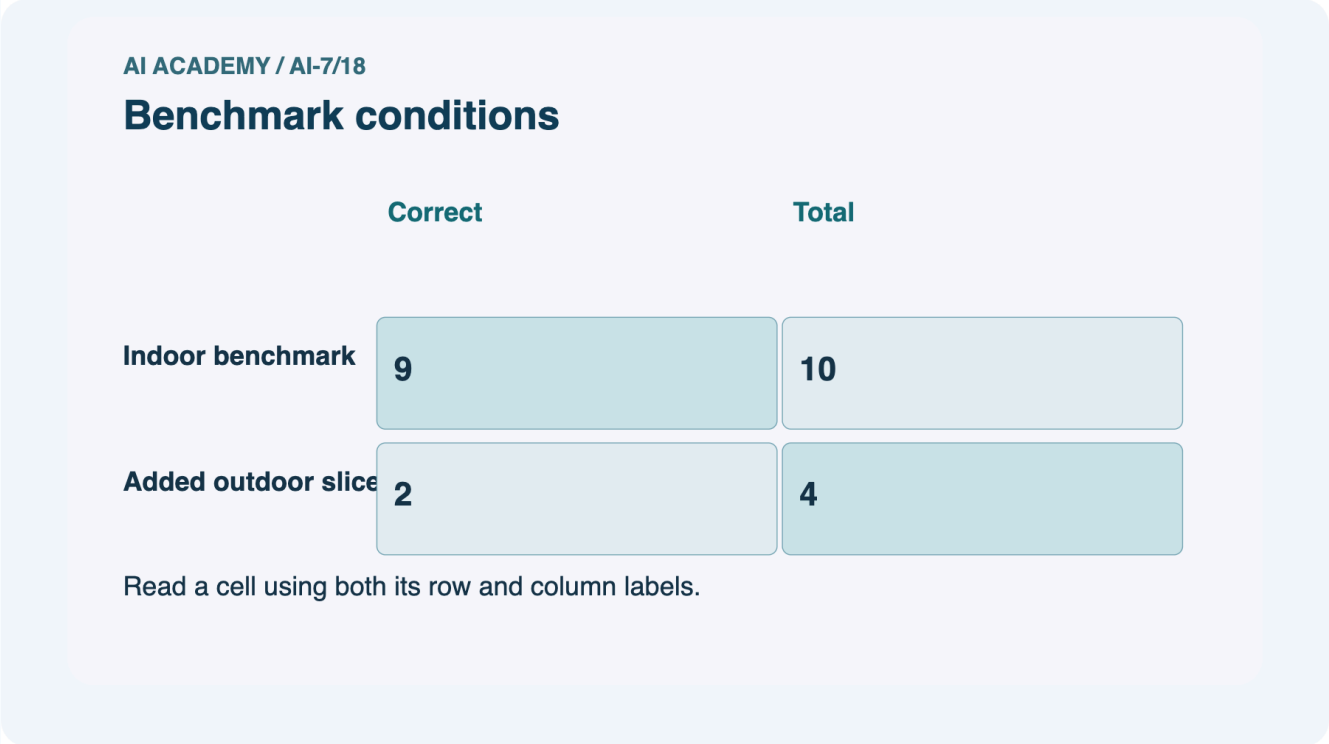
Follow the labelled picture with the matching explanation. Open the vector diagram

Name measured setting: All ten **benchmark** images are clean indoor photos. → Report the result: The observed indoor score is 90%. → Name missing **evidence**: Outdoor shade performance remains unmeasured.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Benchmark conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

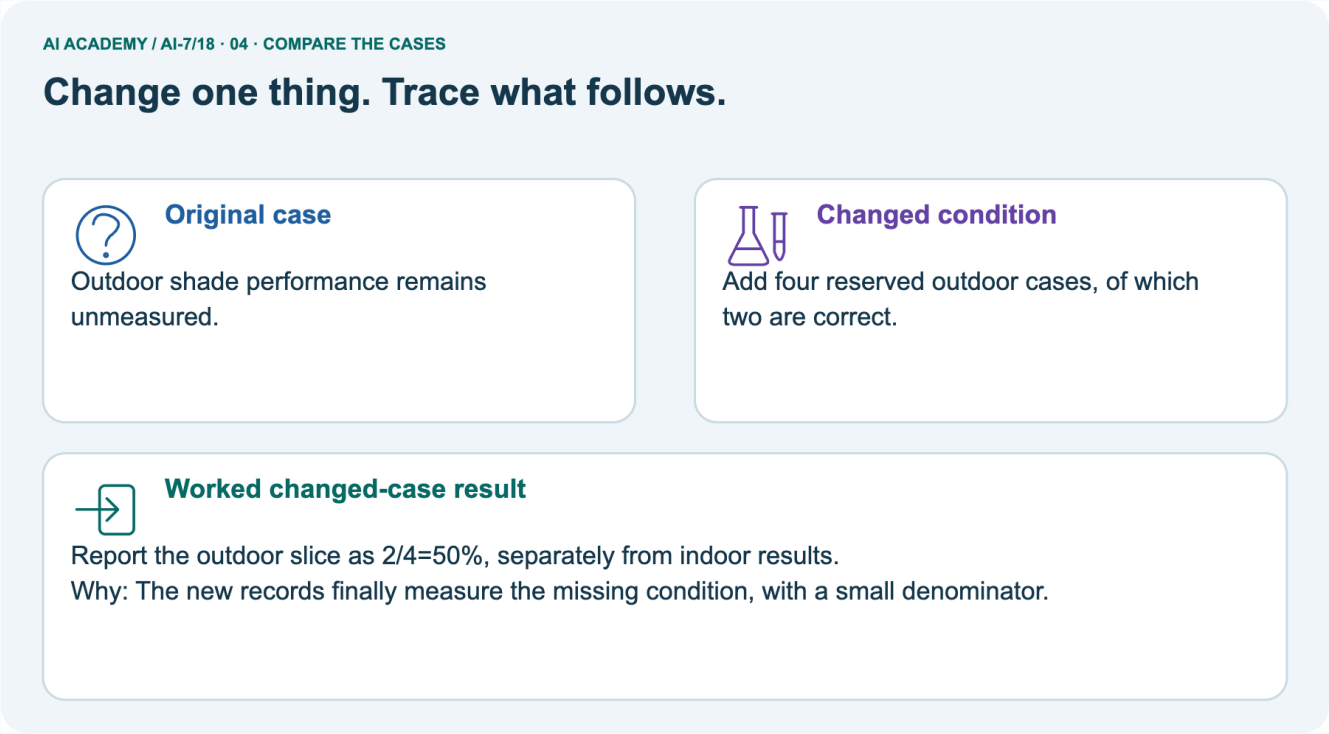
Benchmark conditions. Indoor **benchmark**: Correct = 9, Total = 10. Added outdoor slice: Correct = 2, Total = 4.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Outdoor shade performance remains unmeasured. Changed condition: Add four reserved outdoor cases, of which two are correct. Expected result: Report the outdoor slice as 2/4=50%, separately from indoor results.

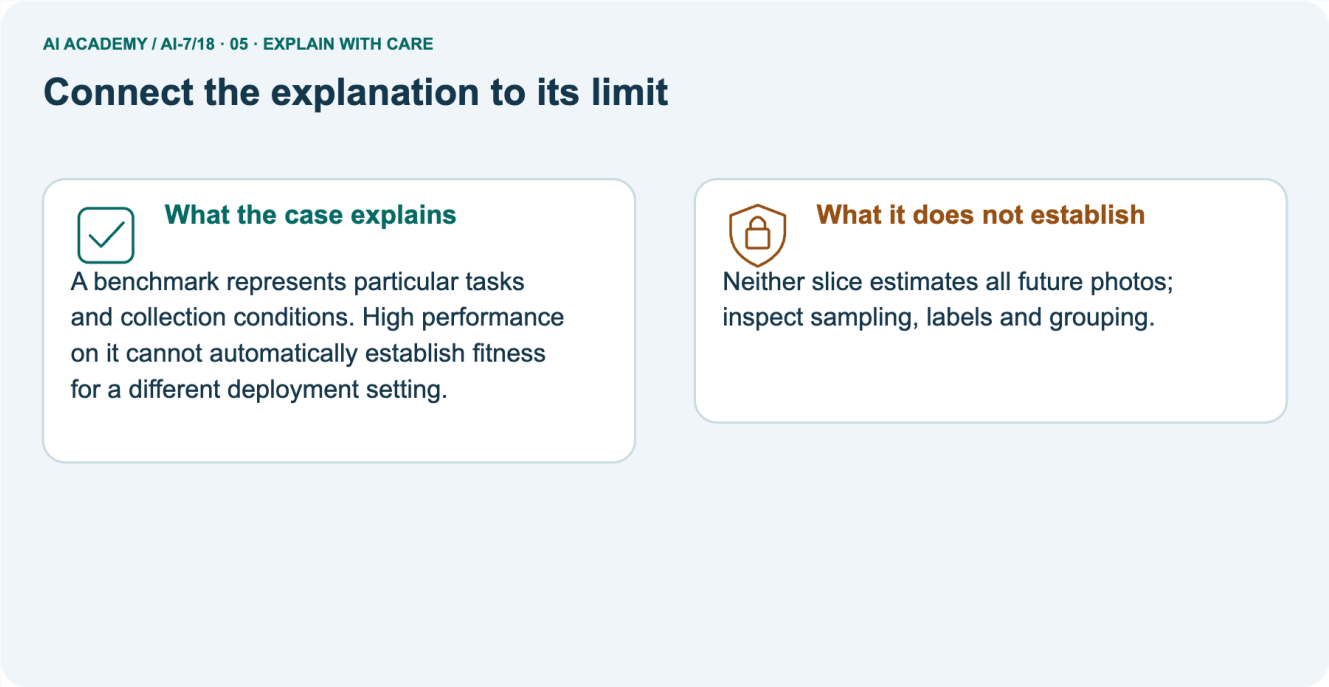
Reason: The new records finally measure the missing condition, with a small denominator.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The new records finally measure the missing condition, with a small denominator.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **benchmark** represents particular tasks and collection conditions. High performance on it cannot automatically establish fitness for a different deployment setting.

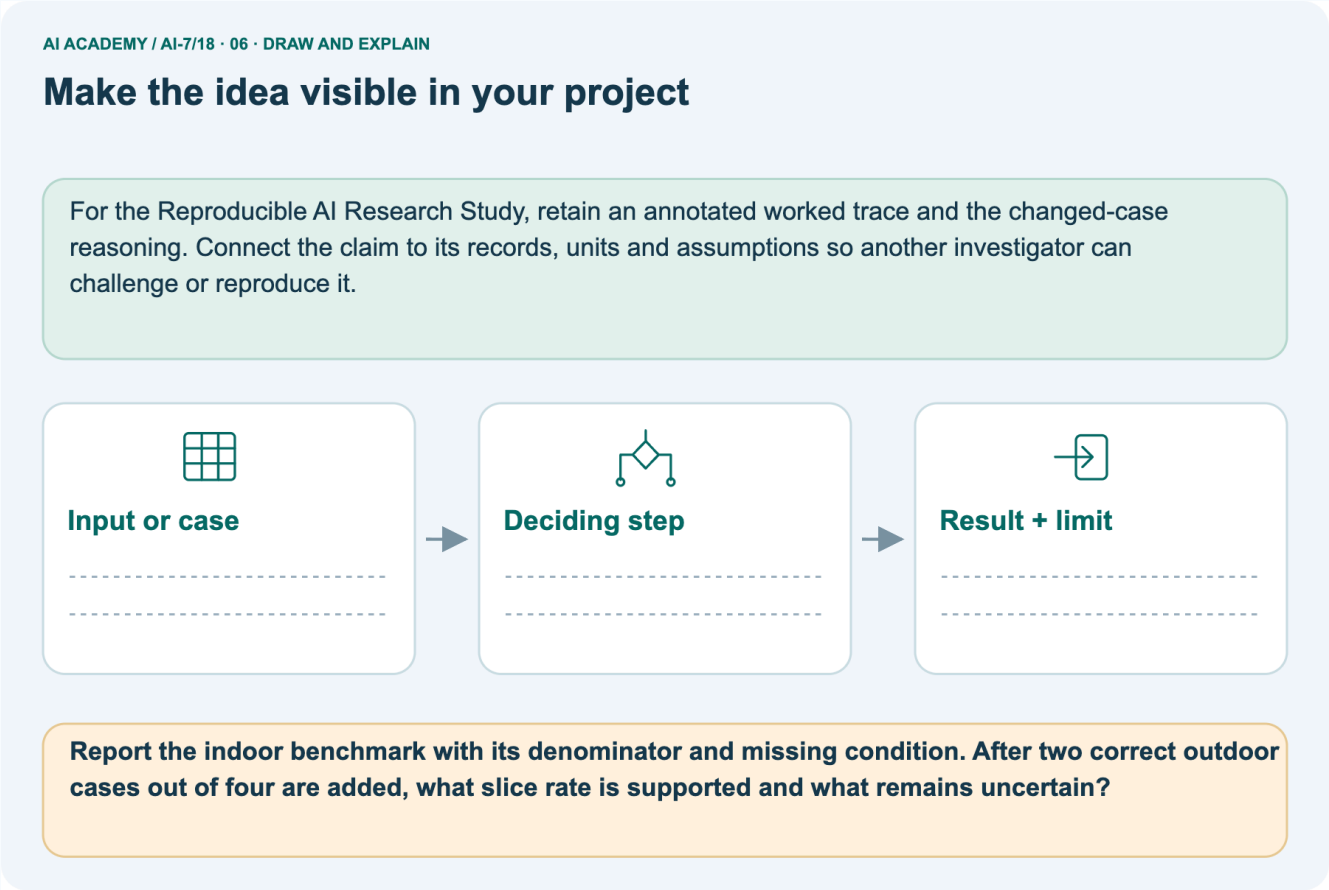
Limit: Neither slice estimates all future photos; inspect sampling, labels and grouping.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Neither slice estimates all future photos; inspect sampling, labels and grouping.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Report the indoor **benchmark** with its denominator and missing condition.

After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: All ten **benchmark** images are clean indoor photos. The observed indoor score is 90%. Outdoor shade performance remains unmeasured.

Changed case: Report the outdoor slice as 2/4=50%, separately from indoor results. The new records finally measure the missing condition, with a small denominator.

▼ **Earlier wording, alternatives and picture transcripts**

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Dataset and benchmark limitations: the case

02. Follow the deciding steps

03. Benchmark conditions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.